

RePlate: A User-Driven Solution for Empowering Sustainable Food Consumption in University Dining

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Abstract

RePlate is the first food reviews app to foreground sustainability in institutional dining services, enabling university students to make carbon-conscious food choices and minimize food waste while tracking sustainability. An estimated 170 million metric tons of carbon dioxide is produced annually by food waste disposal in the United States. University dining halls are a high contributor due to their production and distribution for large student populations. Consulting with representatives at the University of California, Los Angeles (UCLA), we identified the lack of student engagement and manual data collection as pain points in the current food waste stream. Research on food waste interventions, a user survey with 24 respondents, and three in-depth user interviews provided insights into how food reviews and quantitative tracking could minimize food waste on campus. RePlate has the potential to be scalable to other dining scenarios in promoting individual sustainability and utilizing aggregated data for menu development.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**;

Keywords

Sustainability, Food consumption, Food waste reduction, User-centered design, Technology

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1 INTRODUCTION

Having a regular food source is essential to our health and well-being; however, with modernized food processing and distribution comes excessive waste from production to unfinished meals. The Environmental Protection Agency estimates that the annual food loss and waste in the United States produces greenhouse gas emissions equivalent to 170 million metric tons of carbon dioxide [3]. Environmental detriments begin at the stage of agricultural production of resource-intensive foods. On average, 446 grams of greenhouse gas emissions are produced per 100 grams of meat products during agriculture [5]. Food waste has a compounding effect with the daily occurrence of food consumption, particularly in large-scale dining such as university dining halls. The average American household throws away about 6.2 cups of food per week and about 0.940 ounces per person [9]. In comparison, food waste audits from the University of California, Los Angeles (UCLA) reveal an average of 150-250 pounds of edible food wasted per meal per dining hall, or 2.5 ounces of edible food waste per person [13]. Although there are surplus food sources being disposed of, food access issues remain prevalent. A survey of 22 postsecondary schools in the United States found that 44% of participants were considered food insecure [10]. The co-occurrence of surplus food disposal and food insecurity highlights the need for more equitable allocation of food resources to combat overconsumption [6].

User research revealed that dining hall food waste often starts with students' food choices, with food waste typically occurring when students try new dishes that end up differing from expectations. User interviews also suggested that dining hall information is insufficient for students to make sustainability-informed decisions.

Our solution, RePlate, is a menu reviews digital app that enables and empowers students to make sustainable food choices through user-generated reviews and sustainability resources. RePlate utilizes carbon calculations for each dish to help users identify low-carbon footprint food options. RePlate also includes surplus food resources to divert food waste and combat food insecurity. This solution enables students to track and manage sustainable behaviors with user food reviews, self-reported food waste, and calculated cumulative carbon emissions consumed during meals.

RePlate supports the United Nations Sustainable Development Goals (SDGs) towards *Goal 13: Climate Action* by reducing carbon emissions produced by food waste disposal. It supports *Goal 2: Zero Hunger* by promoting food recovery programs that redirect

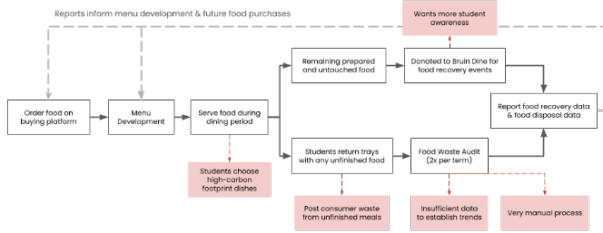


Figure 1: Gap Analysis of the Food Waste Prevention Process at UCLA.

resources to food-insecure populations. Insights from user reviews and sustainable tracking features inform and help users choose sustainable options, manage food waste, and promote sustainable consumption and production behaviors, supporting *Goal 12: Responsible Consumption and Production*.

2 BACKGROUND

2.1 Current State

To discover opportunities to reduce food waste in universities, we interviewed the UCLA Dining Sustainability Representative and the Club President of Bruin Dine, a student-run food recovery program. We conducted informational interviews with these stakeholders to learn about food waste reduction tools and processes. Currently, post-consumer food waste is evaluated through food audits that analyze food waste at tray return stations; the analysis then informs dining hall menu development. However, since the food audit process is done manually, it only occurs twice in a school term, which has been insufficient to establish trends. Alongside tracking food waste, stakeholders also wanted to increase student engagement to prevent food waste. In Figure 1, a gap analysis outlines the current state of food waste prevention efforts at UCLA, highlighting the pain points stated by stakeholders. Stakeholders also want to promote food recovery resources to divert food disposal and reduce food insecurity on campus. Given these goals, the design aimed to increase student engagement in sustainable practices.

2.2 Prior Work

Previous studies have explored why food waste occurs in university dining halls and how to increase student involvement. A multi-campus study by Wiriyaphanich et al. analyzed food choice and waste behaviors by surveying students on the level of certainty that they would enjoy their selection before a meal and then analyzing the food waste after the meal. Higher certainty in the student's food choice was correlated with less food waste, indicating that increasing certainty in food choice could reduce food waste [11]. Alattar et al.'s survey conducted at Portland State University had 55% of students report taste preferences as a top reason for food waste and 31% reporting overestimating portion size as their top

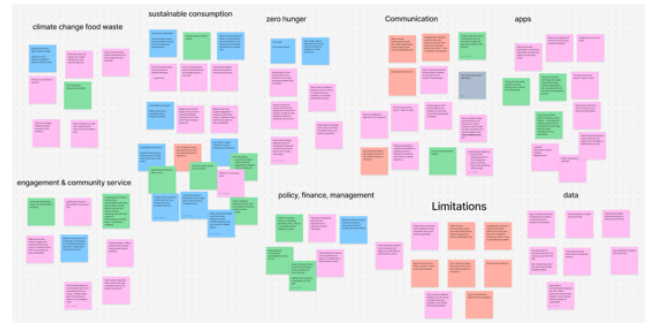


Figure 2: Affinity diagramming and thematic analysis for solution ideation.

reason [1]. Results from Wiriyaphanich and Alattar's studies suggest supplementary information is needed for students in deciding on appropriate food selections that match their preferences in efforts to reduce food waste. Online food review platforms are often utilized to increase decision certainty, as demonstrated in Cho & Chan's study, where participants on average strongly agreed that online reviews impacted their decision-making [4]. Alongside user-generated reviews for qualitative information, digital platforms have the opportunity to promote sustainability to drive behavioral changes. Baungaard et al. identified a lack of education on implementing sustainable diets as a barrier to sustainability for university students, suggesting students are unable to distinguish which food options are more sustainable than others [2]. Quantifying sustainability was found to help users discover sustainability opportunities, as demonstrated in Hoffman et al.'s study, where participants' carbon footprint decreased after the use of Carbon Footprint Tracking Apps [7]. Digital platforms could also enable students to discover surplus food options, both as a practice to divert waste and to raise awareness of food insecurity resources. Krupp & Lebo exemplified the effectiveness of surplus food platforms through their campus food recovery app, which redirected over 1,000 food items to students rather than to the landfill during the trial period [8].

Combining insights from stakeholder interviews and previous studies, the team conducted thematic analysis using an affinity diagram to ideate solutions (see Fig. 2). In response to the manual data collection currently required for food waste audits, RePlate was designed as a solution for food waste to be tracked by the students while also increasing engagement in sustainability. Leveraging the food waste reduction methods explored in previous studies, RePlate utilizes food reviews to both prevent food waste and quantitatively track sustainability.

3 User Research

To understand user food choices and sustainability practices within our sample, we released a user survey asking students about their typical dining hall experiences and food waste habits. The survey consisted of 24 participants who have had dining experience and owned a dining hall meal plan at UCLA in the past two years. The survey results validated reasons for food waste found in previous studies, with top reasons being only liking some parts of a dish,

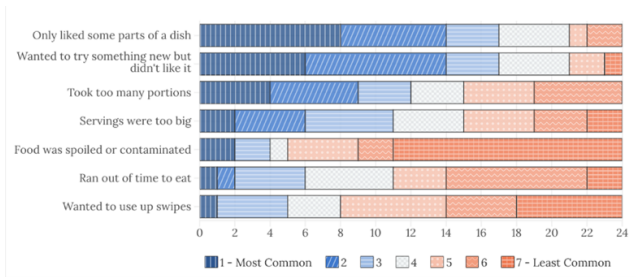


Figure 3: Reasons for Food Waste in UCLA Student Dining, Ranked from Most Common to Least Common.

wanting to try something new but not liking it, and taking too many portions (see Fig. 3).

A total of 87.5% of respondents reported that they received food recommendations from friends. When asked about factors for choosing dishes to eat, respondents reported “food reviews & recommendations” as the top reason. Respondents also reported that sustainability was the least important factor in choosing dishes in their current dining experiences, and only 4.2% of participants utilize surplus food resources.

Based on the insights revealed in the user survey, user interviews were conducted with three participants to gain a deeper understanding of users’ mindsets behind user decision-making and sustainability knowledge. The cognitive walkthrough method was used to observe users’ current process for choosing a dish from the menu and their sustainable choices. When selecting a dish, participants would typically look at the ingredient list or try to look for pictures to better evaluate whether or not they would enjoy the dish. When asked to describe the last time they had food waste, students explained that the dishes turned out to be unexpectedly filling or spicier than they had anticipated when choosing the dish and they were unable to finish. In terms of sustainable choices, users explained that from the menus alone, they were unable to distinguish what made some dishes more sustainable than others. Students also reported on their food tracking behaviors, often storing pictures of their meals on social media or their personal devices to remember meals they enjoyed and get the same dishes again.

A user persona was crafted to represent the typical user’s current sustainable practices and guide feature design to solve for the user pain points that lead to food waste (see Fig. 4).

4 Design Process

4.1 Prototyping

The team used the Crazy 8s method as a quick iterative way to begin low-fidelity wireframes to establish the user flow according to insights from the user survey and interviews. RePlate’s design system was created using the wireframing and prototyping tool, Figma, to standardize visual elements and produce a high-fidelity prototype for user testing.

4.2 Usability Testing and Feedback

Two rounds of evaluation were conducted, with five UCLA students participating in usability testing and 15 students participating in

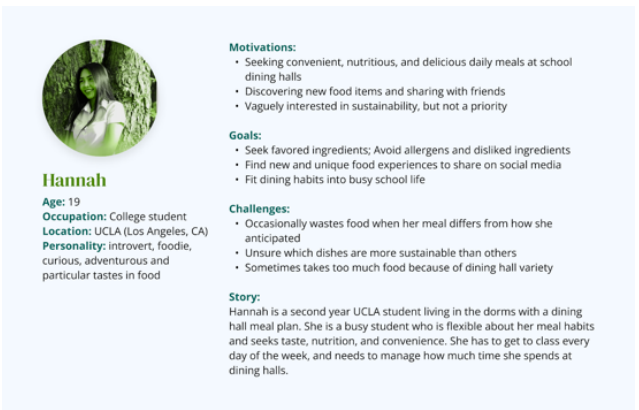


Figure 4: User Persona for the typical student that has meals at the dining halls.

the second round of user evaluation. In the usability testing sessions, the users did cognitive walkthroughs, completed tasks in the prototype (see Figure 5), and provided feedback. The second round of evaluation was conducted remotely with students across eight college campuses across the US to gauge user feedback in a wider scope.

For the usability testing sessions, users conducted three tasks, and they were able to complete the task successfully.

Task 1: Identify the most sustainable and least sustainable menu items. Participants were successful in Task 1 but elaborated that they thought the carbon scores were too precise, which increased the time taken to compare menu items. To improve scannability for users to compare sustainability, a middle tier of carbon scoring was added with a yellow color and with no icon so users could have another visual distinguisher between items.

Task 2: Create a review for a meal consisting of two dishes. In Task 2, participants would miss selecting the second dish and try to progress to the rating process before completing the dish selection, clarifying they expected the process to be selecting and rating dishes one at a time. To signify to the user that all dishes need to be selected at the start, dynamic text was added to the call-to-action button to inform the user how many menu items they are rating. The chip components that show selected menu items were moved closer to the call to action button to further suggest the menu items are multiselect options.

Task 3: Answer questions on their comprehension of a user’s weekly carbon analytics. For Task 3, users reported taking longer to interpret the analytics because the graph was visually dense, and that perception of sustainable improvement is limited from the data visualization alone. Incorporating user feedback, the data visualization was updated to visually separate elements using bars and line graphs, and widgets were provided to help interpret the user’s carbon score.

In the second round of user evaluation, users engaged in an exploration of the interactive prototype and provided their feedback. Students reported that the top three reasons they would use RePlate are to “discover highly rated food,” “discover dishes with lower carbon footprint,” and “share food content with friends.” When asked

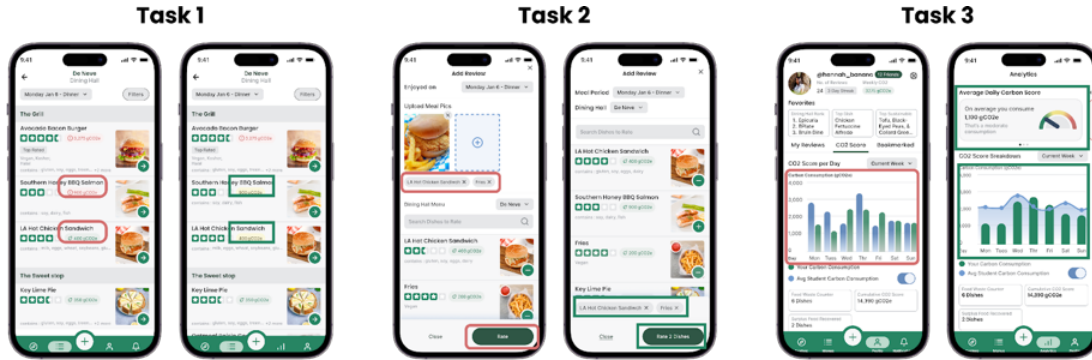


Figure 5: Usability feedback is outlined in a rounded red box on usability testing screens (left) and design iterations are outlined in an unrounded green box on design iteration screens (right) for Task 1, 2, and 3.

about their impression of the App, a user reported that “it helps tailor reviews, recommendations, and even analytics in a way that makes it a bit like social media.” Another user reported, “I liked the focus on food waste and general environmental sustainability.” Some users suggested that the eco food choices could be further highlighted. Additionally, participants mentioned that, for them to adopt the app, bridging it with existing campus dining apps, ensuring that food items are synced correctly to reflect actual options on the menu, and aligning with school-specific dining habits will be very important. Overall, there was a positive sentiment towards the concept and functionalities of the app.

4.3 Stakeholder Evaluation

According to the Sustainability Manager of Housing and Hospitality Services at UCLA, RePlate matches previous proposals for collecting student feedback through software, and will be able to collect data points that can successfully supplement or even replace waste audits to track food waste more efficiently. With the help of RePlate, dining halls can have more confidence in testing the popularity of new menu items that alleviate material cost, waste, and sustainability issues. Food waste data and student preferences collected (in the form of reviews) could be utilized to inform menu development and prevent food waste starting at the early stage of food production. Some concerns they listed were integration with current menus and student adoption, which aligns with feedback given by our student participants. Overall, they believe Replate can help mitigate food waste from both dining and production.

5 Final Features

Usability testing refined the feature set defined from user research and prior studies. The following features were validated to drive student behavioral change toward sustainable food consumption. The design for RePlate’s main screens can be seen in Figure 6.

- **Discover:** Users can find reviews and increase certainty in their food choices to minimize waste. The explore and menu pages enable users to discover sustainable foods and food access resources like surplus food options.
- **Track:** Reviews made by users act as tools to track their sustainable behaviors, such as cumulative carbon score, food

waste tracking, and tracking food recovery that diverts from landfill waste.

- **Analyze:** The Analytics page provides visualizations of the user’s carbon consumption throughout a time frame (e.g. a week) compared to average student consumption on campus. Data insights identify opportunities for students to increase sustainability.

RePlate provides an experience for students to record and share their food experiences with others while discovering new sustainability options and tracking their habits. Along with the benefits of driving sustainable adoption in students, Replate’s value also lies in its potential for informing sustainable tracking and management in college administration. The data could be integrated with an admin-side dashboard that aggregates student reviews, self-reported food waste, and CO2 consumption across campus, as shown in Figure 7 with a preliminary prototype. The RePlate admin dashboard would utilize insights on dishes to inform menu development, further supporting Goal 13, Sustainable Consumption, by optimizing menus to offer dishes that are least likely to be wasted.

6 Discussion and next steps

Debatably, individual food waste could still be present despite using RePlate. However, tracking student preferences can prevent food waste upstream during food production and menu development. User drop-off is another potential concern, as novelty effects could diminish over time with the effort required to log reviews. Concerns for user drop-off could be mitigated through campus-wide food waste prevention campaigns, which was shown to be effective in Yu et al.’s study that acquired 220 users for their food waste tracker app during a two-week food-saving campaign [12].

Beyond the design itself, it is important to understand how to foster adoption and a dining culture surrounding RePlate, while ensuring the display of all available food items, as raised through feedback. Additionally, RePlate can be further validated by engaging even more stakeholders, such as other university administrators, food service providers, and policymakers. RePlate’s development would benefit from featuring a long-term pilot program, testing its functionalities across different universities, with proper modifications in place tailored to each university to test behavioral impacts, scalability, and reliability. Nevertheless, feedback shows

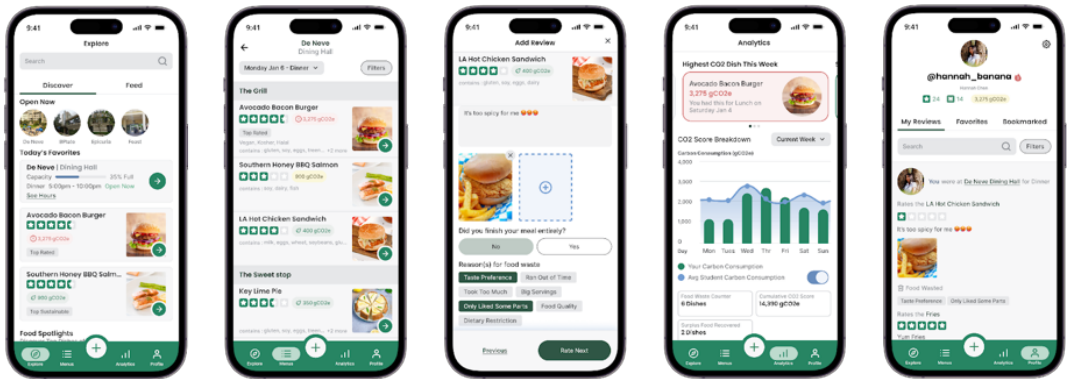


Figure 6: RePlate’s main screens from left to right: Explore, Menus, Add Review, Analytics, and Profile.

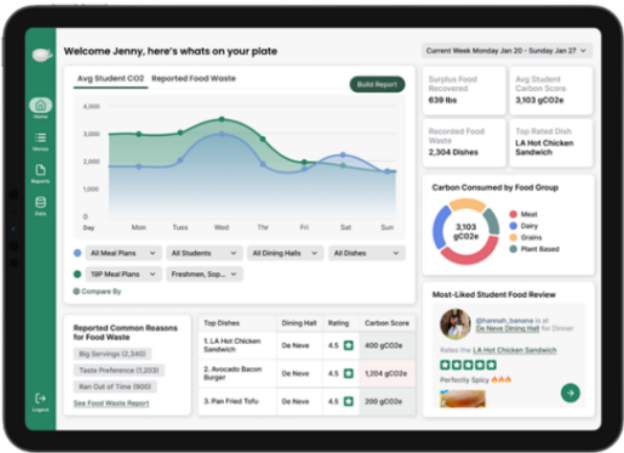


Figure 7: Preliminary prototype of RePlate’s admin dashboard.

that RePlate is likely scalable to other universities in the United States with similar student dining scenes.

RePlate could be scaled to cover other large-scale dining use cases, such as commercial and hospitality dining. RePlate’s unique capability of combining menu reviews with sustainability efforts could be integrated into current food ordering platforms and dining check-out experiences. RePlate’s features are incentivizing for both environmental benefits and potential cost savings when utilizing direct insights for menu development. This solution addresses UN Sustainable Development Goals as a software that enables users to utilize reviews to choose preferred dishes and reduce food waste (SDG 13). Dish carbon scoring and weekly aggregated analytics empower users to track sustainability and discover opportunities to reduce their carbon footprint (SDG 12). RePlate promotes surplus food programs, increasing resource visibility to users facing food insecurity (SDG 2). RePlate demonstrates high potential as a product due to the sustainability-driven values behind its features, validated by prior research and user testing.

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